Szczepan Letkiewicz

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SUMMARY

Second-year Dual Master's student in Aerospace Engineering and Wind Energy, with strong analytical skills in fluid dynamics, computational modeling, and experimental testing. Experienced in independent research, collaborative projects, and physics-based modeling to solve complex engineering problems.

EDUCATION

Master of Science Aerospace Engineering
Sep 2024 - Jul 2026

Delft University of Technology (TU Delft), The Netherlands

Average grade: 8.71/10

- **Specialisation in Aerodynamics**
  - Relevant knowledge: Flow Measurement Techniques, Experimental Methods and Simulation, Fluid Flow Data Processing and Visualization, Fluid-Structure Interaction, Numerical Methods, Multidisciplinary Design Optimization.
- **Honours Programme,** 20 ECTS | Top 10% Selective Extracurricular Programme
  - Assisted in an academic research project on eVTOL aerodynamics for urban air mobility.

Master of Science Wind Energy
Sep 2024 - Jul 2026

Technical University of Denmark (DTU), Denmark

Average grade: 11.2/12

- **Specialisation in Aerodynamics**
  - Relevant knowledge: Turbulent Flows & Turbulence Modeling, Computational Fluid Dynamics (CFD), Advanced Fluid Mechanics, High-Performance Computing (HPC), Programming.

Bachelor of Science Advanced Technology
Sep 2021 - Jul 2024

University of Twente (UT), The Netherlands experiance

Average grade: 7.94/10

- **Specialisation in Aeronautics**
  - Relevant knowledge: Control Theory, System Dynamics, Signal Analysis & Filtering, Data Statistics & Probability.
- **Honours Programme,** 30 ECTS | Top 10% Selective Extracurricular Programme

EXPERIENCE

Master Student Research
Oct 2025 - Jul 2026

Von Karman Institute for Fluid Dynamics (VKI), Belgium

- Conducting LES and URANS simulations to train multi-fidelity neural network surrogate models for flow prediction.
- Combining simulation data with theoretical modeling for efficient aerodynamic and aeroacoustic analysis.
- Applying machine learning for flow reconstruction, multi-fidelity modeling, and uncertainty quantification.
- Attending VKI lecture series including:
  - **Experimental Fluid Mechanics:** PIV, Laser Doppler Velocimetry, Flow Visualization and Optical Measurement.
  - **Machine Learning for Fluid Dynamics:** PINNs, reinforcement learning, data assimilation, and digital twinning to enhance AI-driven experimental diagnostics.

UPCOMING PUBLICATIONS

- SZ. Letkiewicz, "Semi-Analytical Modeling of Blade Wake Deficits for Propeller–Wing Interaction in Aeroacoustics," to be presented at the 32nd AIAA/CEAS Aeroacoustics Conference, May 2026. (in preparation)
- Example title: "Multi-Fidelity Surrogate Modeling of Propeller–Wing Wake Interactions for Aeroacoustic Prediction", based on ongoing Master's research at VKI, intended for journal submission. (in preparation)

SKILLS

Relevant Software: Python - MATLAB - C++ - OpenCV (Basics) - Linux / Bash - Git - ParaView - STAR-CCM+

Languages: Swiss German (Native), Polish (Native), English (C2), German (C1)